

WASP-52b

R-1522

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_191015 · created 2026-08-01T17:19:05+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458771.801 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.213 +/- 0.062
Transit depth (Rp/Rs)^2	4.55 +/- 2.66 [%]
Orbital Inclination (inc)	87.28 +/- 2.79
Airmass coefficient 1 (a1)	0.97 +/- 0.51
Airmass coefficient 2 (a2)	-0.12 +/- 0.25
Scatter in the residuals of the lightcurve fit is	50.5 %
Best Comparison Star	#3 - [180, 441]
Optimal Aperture	4.41
Optimal Annulus	21.15
Transit Duration (day)	0.083 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.213 ± 0.062 vs 0.1646 (deviation 0.71σ)
Duration fit vs published	0.083 d vs 0.0758 d (deviation 0.47σ)
Mid-time O–C	-11.8 min
Depth SNR	3.1
Residual scatter	50.5 %

Light curve

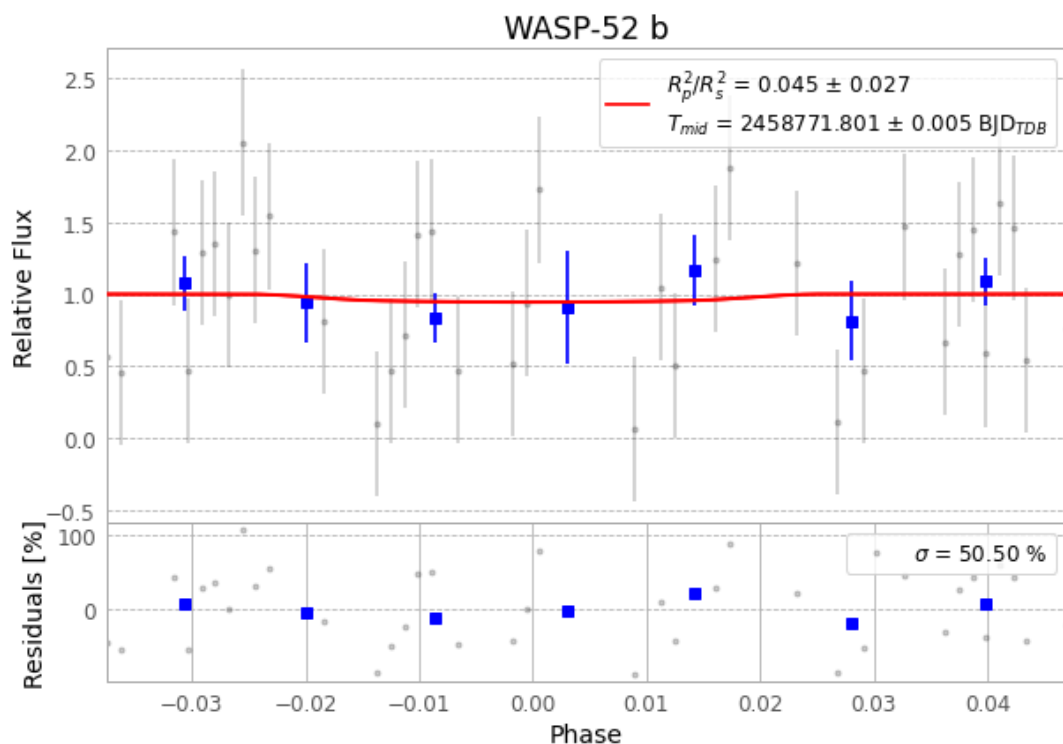


Figure 1 — FinalLightCurve_WASP-52 b_2019-10-14.png (SHA-256 bd6ff0e0a8774ec4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [533, 419], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 377b0111d0d44fe2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-52191015053014.FITS → WASP-52191015091512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-191015.log (12dc1e7fed4c...), FinalLightCurve_WASP-52 b_2019-10-14.png (bd6ff0e0a877...), FinalLightCurve_WASP-52 b_2019-10-14.csv (f04ed9aa15d6...)

Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 17:19Z.